

# BANCO FEDERAL DE FINANÇAS

## Marketing Campaign Analysis

Prepared by North Wind Consulting

### 1 The Spray-and-Pray Tax

In the three years of your campaign, the month of May had the most calls, totaling 12,370 contacts. You lost  $-\$2.61$  on each of those calls. In May you did not hold back at all in your calling and reached out to thousands of people in low-conversion groups that almost always lose money. By contrast, in high-conversion months you called 1,815 people and made  $\$15.58$  per call.

| MAY MASS CALLS               | CAMPAIGN AVERAGE             | TARGETED MONTHS              |
|------------------------------|------------------------------|------------------------------|
| <b><math>-\\$2.61</math></b> | <b><math>-\\$0.43</math></b> | <b><math>\\$15.58</math></b> |

#### Every Call Has a Price Tag

We already know targeted campaigns work. The question is how to scale it.

#### The Spray-and-Pray Tax

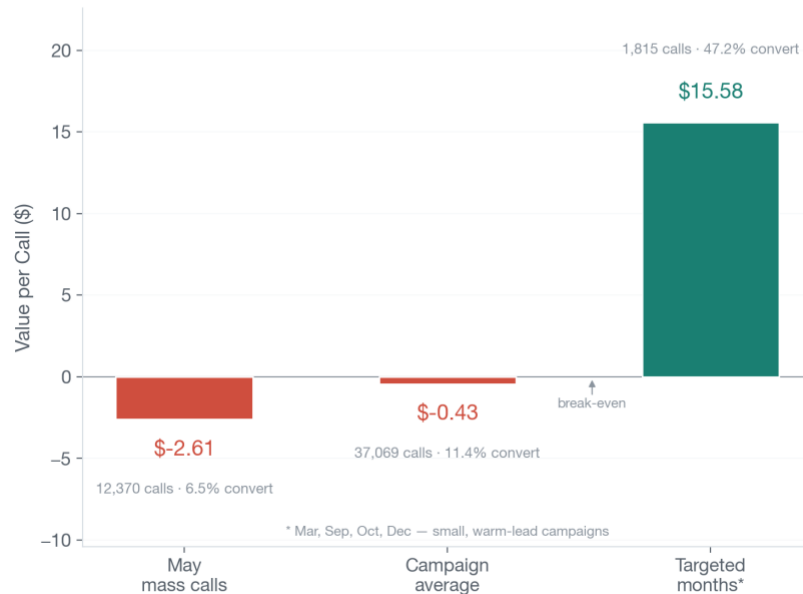


Figure 1. Value per call across May mass calls, campaign average, and targeted months.

A targeted approach focused on high-value demographics is the answer; scaling that approach is the challenge. Our machine-learning (ML) pipeline serves as the bridge between quantity and quality, allowing high-value campaigns to be scaled across tens of thousands of prospects.

## 2 What We Can Do for You

We built our model to help you reach consumers who need your services - and to help you make money. In a recent trial campaign, you called 410 people and lost \$157. Our model would have turned that \$157 deficit into an \$824 profit.

In your next proposed campaign, contacting all 4,100 people on your call list will likely lead to a loss of approximately \$1,600. We project that once running your call list through our model, you will make \$7,300 by stripping your list down to the 480 highest-value prospects.

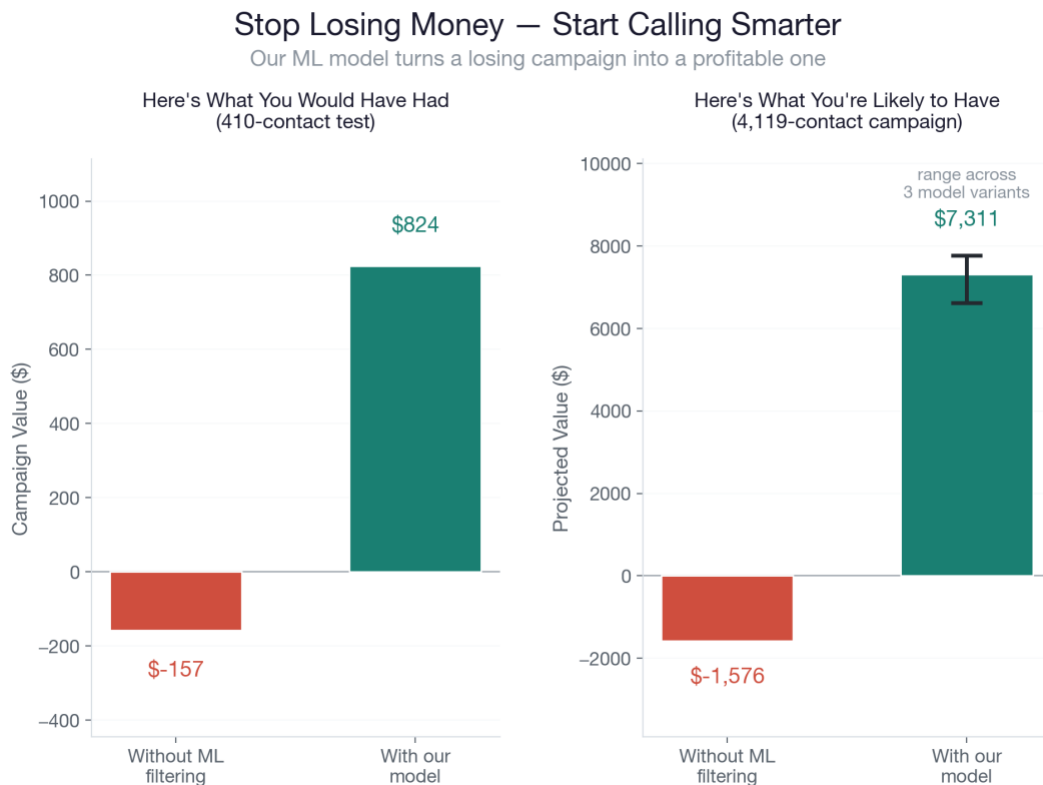


Figure 2. Recent 410-contact test (left) and the projected 4,119-contact campaign (right), with and without the model.

## 3 People to Call — and People Not To

To maximize resource efficiency, our model cleanly separates the database into high-yield priority targets and immediate-skip candidates.

**The Golden List** focuses entirely on our high-conversion groups. At the very top are previously converted customers, who boast a phenomenal **65.8%** conversion rate. Recognizing this massive opportunity, the model shifts this group from a tiny 3% of our historical call list up to 21% of our total outreach.

Similarly, students and retirees exhibit exceptionally strong conversion rates at 31.4% and 25.2% respectively. Retirees are a particularly prime audience; they represent high-liquidity accounts actively

looking to lock in reliable term-deposit yields during fluctuating economic climates. To capture this capital, the model scales our retiree outreach from a baseline of just 5% up to 23%.

**The Black List** identifies the high-waste groups where our budget goes to die. Customers contacted via traditional landlines (5.3% conversion) and those in blue-collar employment blocks (6.8% conversion) drastically underperform our baseline.

Historically, these two segments clogged our pipeline, consuming a staggering 58% of all marketing calls while returning almost no value. The model aggressively prunes this dead weight, shrinking their combined presence on the call list down to a lean 11% so our team stops wasting time on dead ends.

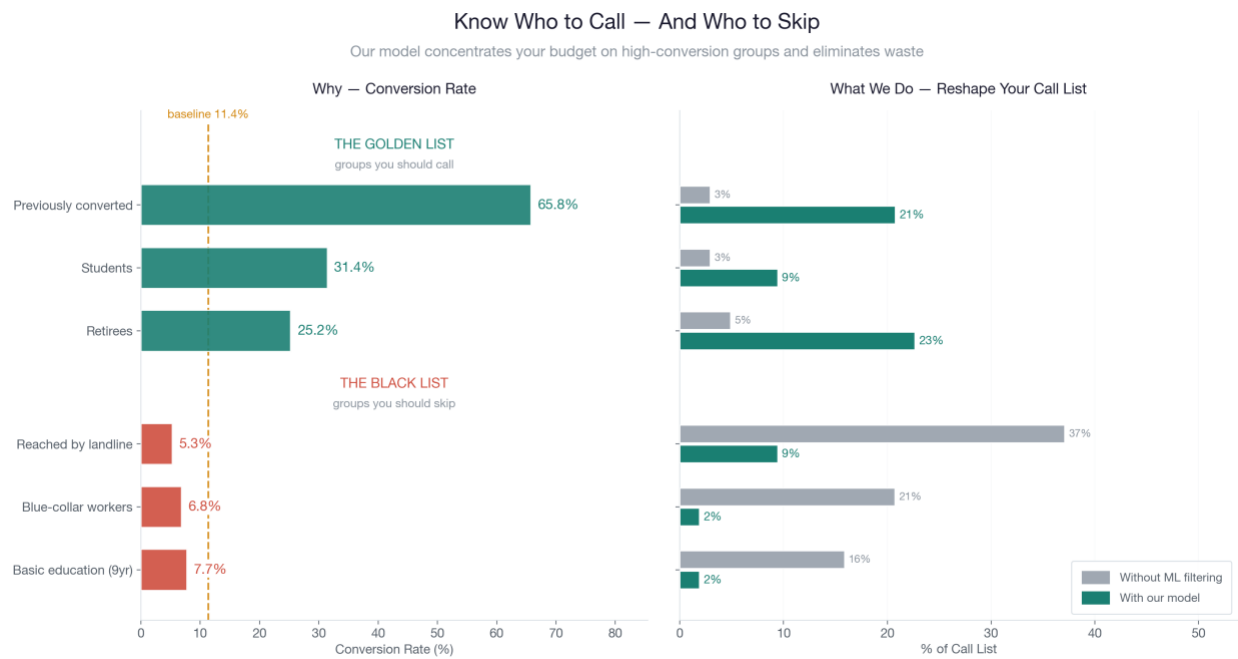


Figure 3. Conversion rates by segment (left) and the resulting call-list reshape (right).

## 4 Steps for Improvement

The best metric we currently have to predict the value of a customer is how likely they are to place money in a CD. To project the cost and value of each call we use the average time on a call (30 min) and the average amount of money placed in a CD (\$4,700). Replacing those defaults with your internal labor and savings data is the single biggest lever for tightening the model's value estimates further.

### APPENDIX · Methodology & Code

Full analysis notebook, model definition, and reproduction steps are available at [nelsbuhreley.github.io/CSE-450-Machine-learning/module\\_2-bank/page/technical](https://nelsbuhreley.github.io/CSE-450-Machine-learning/module_2-bank/page/technical)